LGTC: Local–Global Tri-Consistency Network for Semi-Supervised Change Detection of Remote Sensing Images

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***Abstract*—Recently, semi-supervised change detection (SSCD) has attracted considerable attention due to its remarkable capability to enhance model performance with limited annotated data. However, most existing SSCD methods rely primarily on pixel-level consistency learning to leverage unlabeled data. Although this approach can provide basic prediction consistency constraints, it exhibits notable limitations in capturing spatial continuity and global semantic coherence in structural change regions and is easily affected by noise. To address this issue, we propose a novel SSCD framework, termed the local–global tri-consistency (LGTC) network. LGTC incorporates a tri-consistency learning (TCL) strategy at the pixel, region, and image levels, which provides complementary supervisory signals from fine-grained to global semantic scales, significantly enhancing representation ability and robustness on unlabeled data. Furthermore, a local–global interaction block (LGIBlock) is introduced to integrate local feature details with global context. Within it, the global frequency Transformer block (GFTBlock) employs frequency-domain attention to achieve efficient modeling of global information while effectively suppressing background noise, further enhancing the recognition performance of the model. Experimental results demonstrate that LGTC achieves state-of-the-art performance, attaining *F*1-scores of 87.12%, 88.65%, and 83.25% on the LEVIR-CD, WHU-CD, and CDD-CD datasets with 5% labeled data, respectively, fully verifying the effectiveness of the proposed method. Our source codes are available at https://github.com/sherryxu21/LGTC.**

***Index Terms*—Change detection (CD), remote sensing (RS), semi-supervised learning (SSL), tri-consistency learning (TCL).**

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I. Introduction

CHANGE detection (CD) identifies significant surface changes by analyzing remote sensing (RS) images acquired at different times over the same region [1]. As a key technology in Earth observation, CD has been widely applied in urban planning [2], environmental monitoring [3], disaster assessment [4], and resource management [5].

In recent years, significant progress has been made in fully supervised deep learning-based CD methods [6], [7]. Liu et al. [8] introduced a deep Siamese convolutional network with dual-task constraints (DTCDSCN), which incorporates semantic segmentation as an auxiliary objective and utilizes a dual-attention mechanism to enhance the learning of discriminative features. Transformer structure has been increasingly adopted as an effective feature extraction backbone in CD tasks [9], [10]. For example, to effectively model spatiotemporal context, Chen et al. [11] proposed a bi-temporal image Transformer (BIT) that leverages a Transformer-based encoder–decoder architecture in the semantic domain. To address the characteristics of multitemporal data, Feng et al. [12] proposed a dual-branch multilevel intertemporal network (DMINet), which combines self-attention and cross-attention mechanisms to facilitate feature coupling between bi-temporal images. Recent studies have been actively exploring the potential of alternative network architectures for CD tasks [13], [14]. Notably, the context spatial-aware graph-convolutional network (CSAGC) [15] integrates the complementary strengths of convolutional neural networks (CNNs) and graph neural networks (GNNs) to jointly capture local–global contextual features. Chen et al. [16] pioneered the integration of Mamba into RSCD, designing an innovative state-space model-based encoder–decoder architecture to precisely characterize spatiotemporal correlations. However, these methods exhibit strong dependence on a large number of labeled samples, which are not only costly to obtain but also time-consuming and labor-intensive. On the other hand, the continuous advancement of RS imaging technologies has generated vast amounts of unlabeled data, presenting both opportunities and challenges for CD. Therefore, how to effectively leverage limited labeled samples along with abundant unlabeled data to build models has become a critical research topic in RSCD. Semi-supervised learning (SSL) has emerged as a promising approach to alleviate the dependence on large amounts of labeled data while preserving high detection performance [17].

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Initially, SSL strategies were largely applied to semantic segmentation, leveraging techniques like adversarial learning and self-training to utilize the informative potential of unlabeled samples. For instance, Mittal et al. [18] introduced s4GAN, which employs a generative adversarial network (GAN)-driven adversarial learning strategy to effectively mine discriminative representations from unlabeled data. Based on self-training, Yang et al. [19] proposed ST++ that progressively refines model performance by ranking unlabeled images based on confidence and selectively updating the model through targeted re-training. Consistency regularization stands out as a widely adopted strategy in SSL, showing considerable effectiveness in enhancing model performance [20], [21], [22]. For example, Sohn et al. [23] proposed FixMatch, a streamlined SSL approach, that enforces prediction consistency between weakly and strongly augmented versions of unlabeled images. Building upon this foundation, a unified dual-stream perturbation framework called UniMatch [24] was proposed, which extends FixMatch by incorporating an additional strong view along with a supplementary feature perturbation pathway. Cai et al. [25] introduced an innovative consistency regularization approach grounded in masked image modeling (MIM), where two newly designed MIM-specific strong augmentations are employed to encourage more robust consistency prediction outcomes. These approaches effectively harness the wealth of unlabeled data to enhance the understanding of intrinsic data structures and optimize generalization capabilities.

Currently, numerous studies have investigated the application of SSL in RSCD [26], [27], [28]. Notably, Peng et al. [29] developed the SemiCDNet, a semi-supervised change detection (SSCD) framework built upon GANs, which utilizes two distinct discriminators to promote alignment in both segmentation outputs and entropy distributions across labeled and unlabeled samples. Consistency regularization is a commonly used strategy in SSCD [30], [31]. For instance, Bandara and Patel [32] applied diverse random perturbations to deep difference feature maps and enforced prediction invariance through loss constraints. Zhang et al. [33] proposed a feature prediction alignment (FPA) framework, which explicitly aligns features and prediction maps derived from weakly and strongly augmented RS image pairs. Innovatively, You et al. [34] introduced prototype discriminative learning for CD, which incorporates a feature discrimination loss to improve the compactness within classes and the separability between different classes of prototypes. Unlike conventional pixel-based methods, Zuo et al. [35] introduced an instance-based approach that generates robust pseudo-labels by thoroughly assessing the reliability of individual object instances.

Although these SSCD methods have significantly improved detection accuracy under limited labeled data scenarios, several critical challenges remain unresolved. Traditional consistency learning methods [32], [36] typically rely on pixel-level pseudo-labels to supervise the training of unlabeled data. However, changes in RS images often exhibit patch-wise and structural characteristics. Relying solely on pixel-level supervision makes it difficult to capture the spatial continuity and semantic consistency of such changes, which may lead to false and missed detections. In addition, background noise introduced by factors such as illumination variations and seasonal differences in RS images can also compromise the ability to extract discriminative change features.

To tackle the aforementioned issues, we propose a local–global tri-consistency (LGTC) network for SSCD. As illustrated in Fig. 1, LGTC aims to leverage a large amount of unlabeled data to enhance the detection accuracy and robustness. Specifically, to comprehensively capture complex structural changes in RS images, we propose a tri-consistency learning (TCL) strategy that incorporates multilevel supervision at the pixel, region, and image levels, thereby enhancing the model's capability to learn discriminative change features in unlabeled data. Furthermore, a local–global interaction block (LGIBlock) is introduced into the CD network, which not only effectively extracts local features but also incorporates a frequency-aware attention mechanism to capture global contextual relationships while suppressing irrelevant background noise.

The main contributions of this article are summarized as follows.

1) To effectively capture the complex and multiscale structural changes, we propose a TCL strategy that enforces distinct loss functions at the pixel-, region-, and image-level to achieve multilevel feature alignment and collaborative optimization, thereby significantly enhancing the model's accuracy and robustness in change detection tasks.
2) The proposed LGIBlock consists of the local residual convolution block (LRCBlock) and the global frequency Transformer block (GFTBlock), which enables the model to not only effectively capture the change patterns of different scales but also reduce the interference of background noise on the detection performance.
3) A novel semi-supervised CD framework, LGTC, is proposed, which integrates the TCL strategy with the LGIBlock to effectively model global context, thereby enhancing the model's perception of both spatial continuity and semantic correlations in change regions and its anti-interference ability. Experimental results on three public CD datasets demonstrate its significant performance.

The remainder of this article is organized as follows. Section II provides a detailed explanation of the proposed approach. Section III presents the experiments. Finally, Section IV draws some conclusions of this article and suggestions for future work.

II. Proposed Method

A. Overall Architecture

In SSCD, the training set consists of labeled data $\mathcal{D}_l = \{(x^l_{a,i}, x^l_{b,i}), y^l_i\}^M_{i=1}$ and unlabeled data $\mathcal{D}_u = \{(x^u_{a,j}, x^u_{b,j})\}^N_{j=1}$, where (x^l_a, x^l_b) and (x^u_a, x^u_b) represent the bi-temporal image pairs, a denotes the prechange time, b denotes the postchange time, and y^l represents the ground-truth label. M and N indicate the number of labeled and unlabeled samples, respectively. Typically, N is much larger than M.

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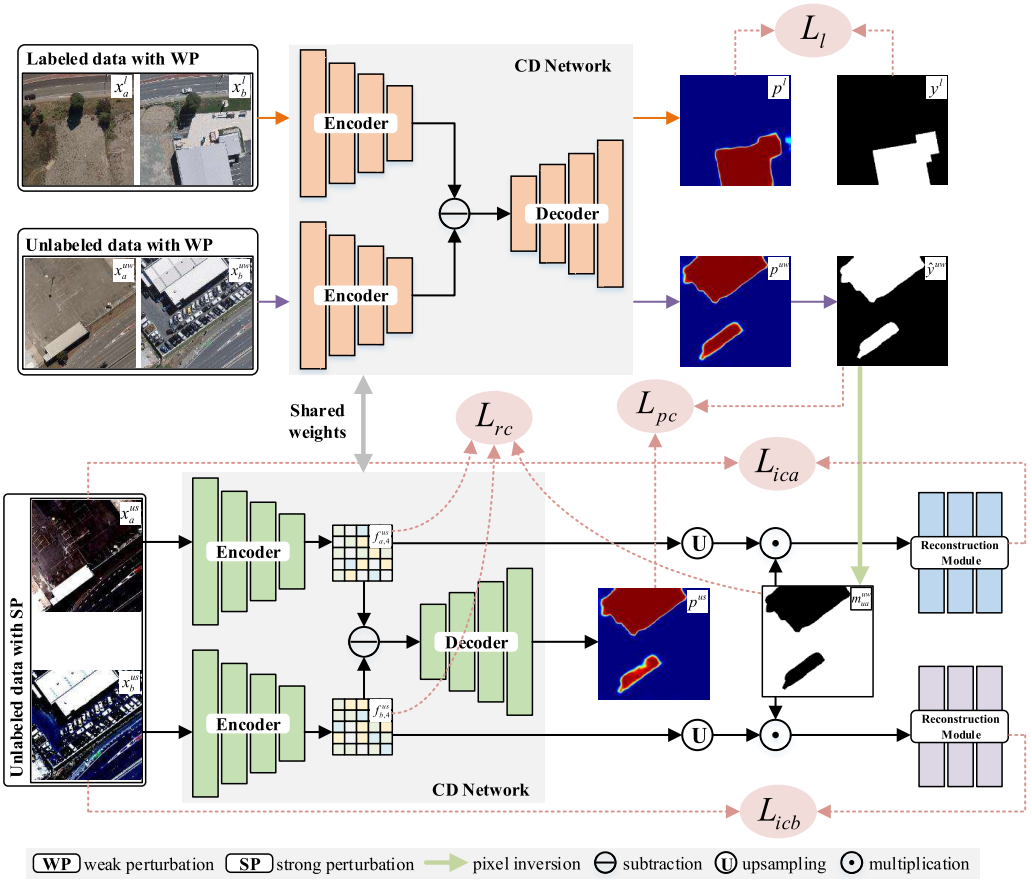


Fig. 1. Overall architecture of the LGTC network for SSCD.

The overall structure of our proposed approach is depicted in Fig. 1. We adopt FixMatch as the foundational framework for end-to-end SSCD, which is based on weak-to-strong consistency regularization. Concretely, each batch consists of B^l labeled image pairs and B^u unlabeled image pairs. For labeled data, weak perturbation (WP) is applied to mitigate overfitting, and the resulting images are fed into the CD network to generate pixel-level prediction maps, which are supervised by manually provided labels. This loss L_l can be formulated as

$$L_l = \frac{1}{B^l} \sum_{i=1}^{B^l} \mathrm{CE}\left(p_i^l, y_i^l\right) \tag{1}$$

where p_i^l is the model prediction of the ith labeled image pair, y_i^l is the corresponding ground-truth map, and $\mathrm{CE}(\cdot,\cdot)$ denotes the cross-entropy [37] loss function.

For unlabeled data, we introduce TCL, which imposes consistency constraints at the pixel-level, region-level, and image-level to capture global structural information. A detailed exposition of this principle will be provided in Section II-D. The unsupervised loss L_u is computed as follows:

$$L_u = L_{\mathrm{pc}} + \alpha \cdot L_{\mathrm{rc}} + \beta \cdot L_{\mathrm{ic}} \tag{2}$$

where L_{pc} denotes the pixel-level consistency loss and L_{rc} and L_{ic} represent the region-level dual-consistency loss and the image-level reconstruction loss, respectively. α and β are their corresponding weights.

To summarize, the overall loss L can be defined as

$$L = L_l + \lambda \cdot L_u \tag{3}$$

where λ is the weight of the unsupervised loss to balance the loss components.

B. CD Network

CD network adopts a commonly used encoder–decoder model [38]. First, the bi-temporal images $(x_a, x_b) \in \mathbb{R}^{H\times W\times 3}$ are fed into the weight-shared encoders to extract high-level semantic features. As illustrated in Fig. 2(a), the encoder first utilizes a projection layer to generate shallow features and then employs a multiscale hierarchical structure to extract deep features, with each stage consisting of multiple stacked LGIBlocks. The LGIBlock integrates two core components, including the LRCBlock and GFTBlock. The LRCBlock is designed to learn the local features, while the GFTBlock incorporates the frequency-domain information to enhance the global feature modeling. These two modules work together to achieve comprehensive feature extraction. The encoder stage can be summarized as follows:

$$f_{a,0} = \mathrm{Proj}\left(x_a\right) \tag{4}$$

$$f_{a,i} = \mathrm{LRC}\left(f_{a,i-1}\right) + \mathrm{GFT}\left(f_{a,i-1}\right), \quad i = 1,2,3,4 \tag{5}$$

where $f_{a,0} \in \mathbb{R}^{H/s_0\times W/s_0\times C_0}$ represents the feature map through the projection layer and $f_{a,i} \in \mathbb{R}^{H/s_i\times W/s_i\times C_i}$ denotes the

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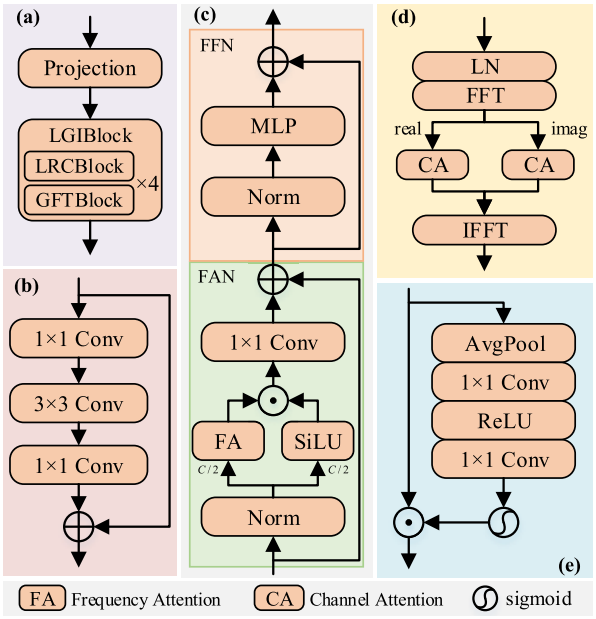


Fig. 2. Architecture of the (a) encoder, (b) LRCBlock, (c) GFTBlock, (d) FA module, and (e) CA module.

feature map extracted from the LGIBlock at the ith stage. s_0 and s_i represent the spatial scaling factors and C_0 and C_i denote the feature dimensions. Proj$(\cdot)$ represents the projection layer, while LRC$(\cdot)$ and GFT$(\cdot)$ denote the LRCBlock and GFTBlock, respectively.

Next, the absolute difference between the bi-temporal feature maps extracted from the encoder is computed to identify potential change regions, followed by the decoder generating the pixel-wise change prediction. The decoder consists of a pyramid pooling module (PPM) [39] and a convolutional upsampling module [40]. The PPM is used to capture the multiscale change information, and the convolutional upsampling module restores the image resolution. The above process can be expressed as follows:

$$p = \mathrm{CUP}\left(\mathrm{PPM}\left(\left|f_{a,4} - f_{b,4}\right|\right)\right) \tag{6}$$

where $p \in \mathbb{R}^{H\times W\times 2}$ represents the change prediction map and 2 denotes the two classes of change and no change. $f_{a,4}$ and $f_{b,4}$ denote the feature maps extracted by the encoder from the prechange and postchange images, respectively. PPM$(\cdot)$ and CUP$(\cdot)$ represent the PPM and the convolutional upsampling module, respectively.

C. Local–Global Interactive Learning

In the LGIBblock, we introduce the LRCBlock and GFTBlock to capture local details and global contextual information, respectively.

1) Local Residual Convolution Block: To effectively learn local patterns and structural information of images, we introduce the LRCBlock to extract the detailed information. As shown in Fig. 2(b), the LRCBlock enhances the local feature representation by stacking multiple convolutional layers and leveraging residual connections. Specifically, the input feature maps are sequentially processed through a 1×1 convolution, a 3×3 convolution, and another 1×1 convolution to achieve the dimension reduction, feature extraction, and restoration to the original dimension, respectively. Then, a skip connection adds the original input to the transformed features to alleviate the common gradient vanishing problem during the training process. The final output x_l of the LRCBlock can be described as

$$x_l = f_c^{1\times 1}\left(f_c^{3\times 3}\left(f_c^{1\times 1}\left(x\right)\right)\right) + x \tag{7}$$

where x denotes the input feature map and $f_c^{1\times 1}$ and $f_c^{3\times 3}$ represent convolution operations with kernel sizes of 1×1 and 3×3, respectively. Each convolution layer is followed by a batch normalization (BN) layer and a rectified linear unit (ReLU) activation function.

2) Global Frequency Transformer Block: To achieve effective global representation learning and mitigate interference from irrelevant noise, we propose the GFTBlock, whose architecture is detailed in Fig. 2(c).

First, the input feature map is processed through a frequency attention network (FAN) to extract discriminative features. Specifically, the normalized features are split channel-wise into two components, including x' and x''. x' is processed by the frequency attention (FA) module to capture the frequency correlation, and x'' undergoes nonlinear activation via the SiLU function. The parallel-processed features are then element-wise multiplied and integrated by a 1×1 convolution to jointly model spatial and frequency information. Subsequently, the processed features are fused with the original input features via a residual connection, yielding the final output x_f of FAN

$$x_f = f_c^{1\times 1}\left(\mathrm{FA}\left(x'\right) \cdot \mathrm{SiLU}\left(x''\right)\right) + x \tag{8}$$

where x denotes the input feature map and FA$(\cdot)$ and SiLU$(\cdot)$ denote the FA module and sigmoid linear unit, respectively.

Next, x_f undergoes feature transformation through a feed-forward network (FFN). Specifically, the feature map is processed sequentially through a normalization layer and a multilayer perceptron (MLP). The normalization operation accelerates model convergence and improves training stability, while the MLP employs an expansion–activation–compression structure to extract rich and compact feature representations. To minimize information loss and enhance gradient flow, a skip connection is subsequently introduced to generate the final output x_g of the GFTBlock

$$x_g = \mathrm{MLP}\left(\mathrm{Norm}\left(x_f\right)\right) + x_f \tag{9}$$

where Norm$(\cdot)$ and MLP$(\cdot)$ represent the normalization layer and the MLP, respectively.

The GFTBlock leverages FA to effectively model global dependencies and separate the low-frequency global background and high-frequency variation details of the image, thereby improving the perception of structural information and providing more discriminative feature representations for noise suppression. As shown in Fig. 2(d), FA first applies layer normalization (LN) to the input features, followed by the fast Fourier transform (FFT) to transform the features from the spatial domain to the frequency domain. Subsequently, the real and imaginary parts of the frequency spectrum are separately processed by the channel attention (CA) module to emphasize

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task-critical frequency components. The enhanced frequency features are then projected back to the spatial domain via an inverse FFT (IFFT), yielding enriched feature representations. The above process can be expressed as follows:

$$\mathrm{FA}(x) = \mathrm{IFFT}\left[\mathrm{CA}\left(\mathrm{FFT}\left(\mathrm{LN}(x)\right)_r\right), \mathrm{CA}\left(\mathrm{FFT}\left(\mathrm{LN}(x)\right)_i\right)\right] \tag{10}$$

where x represents the input feature map and CA$(\cdot)$ and LN$(\cdot)$ denote the CA and LN, respectively. FFT$(\cdot)_r$ and FFT$(\cdot)_i$ refer to the real and imaginary parts of the frequency representation obtained through the FFT, respectively, and IFFT$[\cdot,\cdot]$ represents IFFT.

FA enables the model to selectively focus on discriminative frequency components and filter out redundant information, which is achieved through CA. The structure of CA is shown in Fig. 2(e). First, the feature map is downsampled by average pooling (AvgPool) to extract the global information. Then, two 1×1 convolutions are used to adjust the channel number, fuse features, and generate attention weights, with nonlinearity introduced between the layers. After that, a sigmoid operation is applied to map the attention weights to the range of $(0, 1)$. Finally, element-wise multiplication is performed with the original feature map to select the key frequency components. The above process can be represented as follows:

$$\mathrm{CA}(x) = \sigma\left(f_c^{1\times1}\left(f_c^{1\times1}\left(\mathrm{AvgPool}(x)\right)\right)\right) \cdot x \tag{11}$$

where x represents the input feature map, AvgPool$(\cdot)$ denotes the average pooling operation, and $\sigma(\cdot)$ represents the sigmoid function.

D. Tri-Consistency Learning

As illustrated in Fig. 1, our method fully leverages the abundance of unlabeled RS images by establishing three complementary supervision mechanisms at the pixel, region, and image levels. This not only effectively enhances the discriminative ability to identify cross-scale structural change regions but also significantly improves the robustness and generalization capability.

1) Pixel-Level Consistency Loss: To acquire robust feature extraction capability, we introduce a pixel-level consistency loss that enforces prediction consistency between weakly and strongly perturbed inputs, thereby improving the model's resilience to input variations. Specifically, weak and strong perturbations (SPs) are first applied to the unlabeled data to generate perturbed views (x_a^{uw}, x_b^{uw}) and (x_a^{us}, x_b^{us}), respectively. These are then fed into the CD network to produce prediction maps p^{uw} and p^{us}. The pseudo-label $\hat{y}^{uw} \in \mathbb{R}^{H\times W}$ is derived from p^{uw} as follows:

$$\hat{y}^{uw}(i,j) = \underset{k=\{0,1\}}{\mathrm{argmax}}\, p^{uw}(i,j,k). \tag{12}$$

For a given pixel (i, j), the value of $\hat{y}^{uw}(i, j)$ is assigned as the class index with the maximum predicted probability across all classes, while $p^{uw}(i, j, k)$ represents the predicted probability of class k at that pixel.

Next, to enhance the reliability of pseudo-labels, a threshold τ is introduced to generate a confidence mask $m^{uw} \in \mathbb{R}^{H\times W}$, which can be formulated as

$$m^{uw}(i,j) = \mathbb{1}\left[p^{uw}\left(i,j,\hat{y}^{uw}(i,j)\right) > \tau\right] \tag{13}$$

where $\mathbb{1}[\cdot]$ denotes the indicator function, which takes a logical expression as input and returns 1 if the expression is true, and 0 otherwise. Specifically, for a given pixel (i, j), the confidence mask value $m^{uw}(i, j)$ is assigned 1 if the predicted probability $p^{uw}(i, j, \hat{y}^{uw}(i, j))$ of the pseudo-label class exceeds the predefined confidence threshold τ, and 0 otherwise.

Finally, the pixel-level consistency loss L_{pc} is defined as

$$L_{\mathrm{pc}} = \frac{1}{B^u}\sum_{i=1}^{B^u} \mathrm{CE}\left(p_i^{us}, \hat{y}_i^{uw}\right) \cdot m_i^{uw}. \tag{14}$$

2) Region-Level Dual-Consistency Loss: The ultimate objective of CD is to accurately identify changed regions in bi-temporal images. However, relying solely on pixel-level supervision fails to capture the spatial continuity and semantic coherence of change areas, leading to fragmented predictions or misclassifications. In addition, if the model predictions exhibit high accuracy and the pseudo-labels closely align with the ground-truth changes, the bi-temporal features within pseudo-labeled unchanged regions should demonstrate high consistency, while those within pseudo-labeled changed regions should exhibit significant divergence. To this end, we propose a region-level dual-consistency loss, which simultaneously emphasizes feature consistency in unchanged regions and feature discrepancy in changed regions, thereby achieving more precise change detection.

First, the inherent stability of unchanged regions in pseudo-labels is exploited to promote consistency learning of the feature representations. Specifically, to align with the spatial dimensions of the feature maps, we first expand the pseudo-labels $\hat{y}^{uw}$ generated from the WP branch into a 3-D tensor and apply a pixel-wise inversion operation to derive the unchanged area mask $m_{ua}^{uw} \in \mathbb{R}^{H\times W\times1}$, which can be expressed as

$$m_{ua}^{uw} = 1 - \mathrm{unsqueeze}\left(\hat{y}^{uw}\right) \tag{15}$$

where unsqueeze$(\cdot)$ represents the dimension expansion operation. Then, to learn more robust feature representations, we apply m_{ua}^{uw} to the bi-temporal feature maps output by the encoder in the SP branch, which can be described as follows:

$$\begin{aligned} f_{a,ua}^{us} &= \mathrm{up}\left(f_{a,4}^{us}\right) \cdot m_{ua}^{uw} \\ f_{b,ua}^{us} &= \mathrm{up}\left(f_{b,4}^{us}\right) \cdot m_{ua}^{uw} \end{aligned} \tag{16}$$

where $f_{a,4}^{us}$ and $f_{b,4}^{us}$ represent the bi-temporal feature maps output by the encoder in the SP branch, $f_{a,ua}^{us}$ and $f_{b,ua}^{us}$ denote the feature maps after applying the unchanged area mask, and up$(\cdot)$ signifies the upsampling operation. The feature maps are first upsampled to match the resolution of the mask, followed by element-wise multiplication with the mask to extract the unchanged region features.

To enforce consistency between the bi-temporal features of unchanged regions, we employ the Kullback–Leibler (KL) divergence [41] to quantify and minimize their distributional differences. Given two probability distributions P and Q over the same random variable ξ, the KL divergence from P to Q is defined as

$$\mathrm{KL}(P,Q) = \sum_{\xi=1}^{n} P(\xi)\ln\left(\frac{P(\xi)}{Q(\xi)}\right) \tag{17}$$

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where $\mathrm{KL}(\cdot,\cdot)$ denotes the computation of KL divergence. Then, the unchanged area consistency loss L_{ua} is defined as

$$L_{ua} = \frac{1}{B^u}\sum_{i=1}^{B^u} \mathrm{KL}\left(f_{a,ua,i}^{\mathrm{us}}, f_{b,ua,i}^{\mathrm{us}}\right). \tag{18}$$

Subsequently, we extend the utility of pseudo-labels to changed regions by introducing a constraint that explicitly enforces feature discrepancy between bi-temporal features in these areas. Analogously to the unchanged region processing, a changed area mask $m_{ca}^{uw} \in \mathbb{R}^{H\times W\times 1}$ is first constructed, which can be denoted as

$$m_{ca}^{uw} = 1 - m_{ua}^{uw}. \tag{19}$$

The mask m_{ca}^{uw} is then applied to the bi-temporal feature maps as follows:

$$\begin{aligned} f_{a,ca}^{\mathrm{us}} &= \mathrm{up}\left(f_{a,4}^{\mathrm{us}}\right)\cdot m_{ca}^{uw} \\ f_{b,ca}^{\mathrm{us}} &= \mathrm{up}\left(f_{b,4}^{\mathrm{us}}\right)\cdot m_{ca}^{uw} \end{aligned} \tag{20}$$

where $f_{a,ca}^{\mathrm{us}}$ and $f_{b,ca}^{\mathrm{us}}$ represent the bi-temporal feature maps after applying the changed area mask. The changed area discrepancy loss L_{ca} is then defined as

$$L_{ca} = \frac{1}{B^u}\sum_{i=1}^{B^u} \mathrm{KL}\left(f_{a,ca,i}^{\mathrm{us}}, f_{b,ca,i}^{\mathrm{us}}\right). \tag{21}$$

The greater the discrepancy between the changed regions of the bi-temporal feature maps is, the smaller the value of L_{ca} is. This design effectively enhances the model's discriminative power for identifying true changes.

To mitigate the impact of class imbalance, when computing the region-level dual-consistency loss, we assign different weights to L_{ua} and L_{ca}, which can be formulated as

$$\begin{aligned} \omega_0 &= \frac{\sum_{k=1}^{B^u}\sum_{i=1}^{H}\sum_{j=1}^{W}\left(\hat{y}_k^{uw}(i,j) == 0\right)}{B^u\times H\times W} \\ \omega_1 &= \frac{\sum_{k=1}^{B^u}\sum_{i=1}^{H}\sum_{j=1}^{W}\left(\hat{y}_k^{uw}(i,j) == 1\right)}{B^u\times H\times W} \end{aligned} \tag{22}$$

where $\hat{y}_k^{uw}(i,j)$ denotes the value of pixel (i,j) in the kth pseudo-label within a batch, ω_0 and ω_1 represent the proportion of unchanged (0) and changed (1) pixels, respectively. These proportions are used as adaptive loss weights to balance the contribution of each component in the region-level dual-consistency loss L_{rc}

$$L_{\mathrm{rc}} = \omega_0\cdot L_{ua} + \omega_1\cdot L_{ca}. \tag{23}$$

3) Image-Level Reconstruction Loss: To further learn globally robust feature representations, we introduce an image-level reconstruction loss, which enhances the model's expressive power by incorporating a feature reconstruction task. This loss optimizes the entire image, preserving more complete global semantic and structural information, which helps reduce feature drift caused by insufficient annotation and improves the model's generalization capabilities. Specifically, after applying the unchanged area mask, the bi-temporal feature maps of the SP branch are fed into a reconstruction module to obtain the reconstructed image pair $(x_a^{\mathrm{rec}}, x_b^{\mathrm{rec}}) \in \mathbb{R}^{H\times W\times 3}$, which can be expressed as

$$x_a^{\mathrm{rec}} = \mathrm{RM}\left(f_{a,ua}^{\mathrm{us}}\right), x_b^{\mathrm{rec}} = \mathrm{RM}\left(f_{b,ua}^{\mathrm{us}}\right) \tag{24}$$

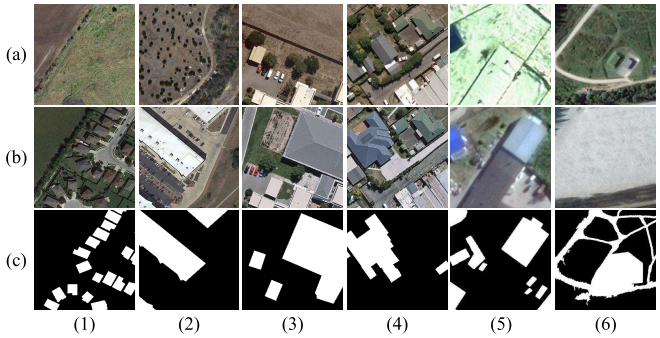


Fig. 3. Examples of LEVIR-CD dataset (1) and (2), WHU-CD dataset (3) and (4), and CDD-CD dataset (5) and (6). (a) RS image at the moment of $T1$. (b) RS image at the moment of $T2$. (c) Ground truth.

where $\mathrm{RM}(\cdot)$ denotes the reconstruction module, which adopts a hierarchical residual structure with three layers. Each layer consists of a 3×3 convolution, a 1×1 convolution, a nonlinear activation, and a residual connection, where the 1×1 convolution is applied before the residual addition to align the channel dimensions. The detailed architecture can be described as follows:

$$x_i = f_c^{3\times 3}\left(x_{i-1}\right) + f_c^{1\times 1}\left(x_{i-1}\right), i = 1,2,3 \tag{25}$$

where x_0 represents the input feature map of the reconstruction module and $x_i (i = 1,2,3)$ represents the output feature map of each layer in the module. This module does not perform spatial upsampling, but restores features through channel-wise transformations.

If the feature representations are sufficiently expressive, the model should be able to reconstruct the original bi-temporal images accurately. Based on this principle, the image-level reconstruction loss L_{ic} is defined as

$$\begin{aligned} L_{\mathrm{ic}} &= L_{\mathrm{ic}a} + L_{\mathrm{ic}b} \\ &= \frac{1}{B^u}\sum_{i=1}^{B^u}\left[\mathrm{mse}\left(x_{a,i}^{\mathrm{rec}}, x_{a,i}^{\mathrm{us}}\right) + \mathrm{mse}\left(x_{b,i}^{\mathrm{rec}}, x_{b,i}^{\mathrm{us}}\right)\right] \end{aligned} \tag{26}$$

where $L_{\mathrm{ic}a}$ and $L_{\mathrm{ic}b}$ represent the reconstruction losses for the prechange and postchange images, respectively, $\mathrm{mse}(\cdot,\cdot)$ denotes the mean squared error loss. The incorporation of the reconstruction error serves to preserve the integrity of the extracted features and improve the generalization performance.

III. Experiments

A. Datasets

We conducted experiments on three publicly available datasets to demonstrate the effectiveness of the proposed method, including LEVIR-CD [2], WHU-CD [42], and CDD-CD [43].

1) LEVIR-CD: The LEVIR-CD dataset consists of 637 high-resolution (HR) Google Earth RGB image pairs with building changes. Each image has a size of 1024×1024 pixels and a spatial resolution of 0.5 m. Due to GPU memory limitations, we processed the image pairs using nonoverlapping sliding windows of 256×256 pixels, resulting in a dataset comprising 7120 training samples, 1024 validation samples,

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TABLE I
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE LEVIR-CD DATASET WITH DIFFERENT LABELED RATIOS

Methods	5%					10%					20%					40%				
	Pre	Rec	IoU	F1	OA	Pre	Rec	IoU	F1	OA	Pre	Rec	IoU	F1	OA	Pre	Rec	IoU	F1	OA
AdvNet	85.18	78.51	69.08	81.71	98.34	86.89	82.33	73.23	84.55	98.58	89.17	83.32	75.66	86.14	98.74	88.51	84.90	76.47	86.67	98.77
s4GAN	87.22	63.61	58.19	73.57	97.84	88.25	65.74	60.45	75.35	97.97	86.33	70.83	63.69	77.82	98.10	85.03	80.57	70.56	82.74	98.41
SemiCDNet	86.56	75.92	67.92	80.89	98.31	87.99	80.71	72.70	84.19	98.57	88.01	84.05	75.42	85.99	98.71	88.74	84.70	76.48	86.68	98.77
RCR	89.82	83.84	76.56	86.72	98.79	89.92	85.18	77.76	87.49	98.85	90.68	85.83	78.87	88.19	98.92	90.36	87.18	79.76	88.74	98.96
RCL	86.63	**86.01**	75.94	86.32	98.71	87.15	86.74	76.90	86.94	98.77	87.54	**87.40**	77.73	87.47	98.82	87.99	**87.93**	78.51	87.96	98.86
FPA	88.43	84.33	75.95	86.33	98.74	88.53	85.94	77.33	87.22	98.81	89.60	86.73	78.79	88.14	98.90	90.89	85.37	78.64	88.04	98.91
UniMatch	**91.10**	81.22	75.25	85.88	98.74	**91.76**	83.03	77.27	87.18	98.85	**92.85**	83.04	78.05	87.67	98.90	89.84	85.80	78.21	87.77	98.87
ECPS	88.88	83.78	75.83	86.25	98.74	90.07	83.77	76.68	86.81	98.80	92.32	82.68	77.36	87.23	98.86	**92.02**	84.90	79.08	88.32	98.94
Sup-only	85.27	79.07	69.56	82.05	98.37	86.53	83.70	74.05	85.09	98.62	89.98	83.99	76.80	86.88	98.80	91.98	84.93	79.08	88.32	98.94
LGTC	90.55	83.94	**77.18**	**87.12**	**98.83**	89.24	**87.07**	**78.80**	**88.14**	**98.90**	90.85	87.39	**80.32**	**89.09**	**98.99**	91.30	87.13	**80.46**	**89.17**	**99.00**

and 2048 testing samples. Representative examples from the LEVIR-CD dataset are illustrated in Fig. 3(1) and (2).

2) Whu-CD: The WHU-CD dataset was captured in the Christchurch area of New Zealand, with a primary focus on building changes. It comprises a pair of aerial images with a spatial resolution of 0.2 m and a size of $32\,507 \times 15\,354$ pixels. To facilitate processing and analysis, the dataset is divided into nonoverlapping subimage pairs of 256×256 pixels. Following this division, the dataset contains 5947 training samples, 743 validation samples, and 744 test samples. Examples from the WHU-CD dataset are presented in Fig. 3(3) and (4).

3) CDD-CD: The CDD-CD dataset comprises HR seasonal change image pairs acquired from Google Earth, with spatial resolutions ranging from 3 to 100 cm/pixel. Unlike the LEVIR-CD and WHU-CD datasets, it encompasses not only building changes but also various other human-induced and natural change types, including vegetation, vehicles, roads, and other landscape alterations. The complete dataset contains 16 000 image pairs, each with a size of 256×256 pixels, divided into 10 000 training pairs, 3000 validation pairs, and 3000 test pairs. Examples from the CDD-CD dataset are shown in Fig. 3(5) and (6).

B. Comparative Methods and Evaluation Metrics

To evaluate the superiority of the proposed LGTC, we conducted comprehensive comparisons with eight state-of-the-art semi-supervised algorithms. These include three semi-supervised semantic segmentation methods: AdvNet [44], s4GAN [18], and UniMatch [24], as well as five SSCD methods: SemiCDNet [29], RCR [32], RCL [36], FPA [33], and ECPS [45].

The change detection performance is evaluated using five quantitative metrics: precision (Pre), recall (Rec), $F1$-score ($F1$), intersection over union (IoU), and overall accuracy (OA). It is worth noting that we exclusively compute and report the change class metrics to emphasize the significance of change detection, where higher values indicate better performance for all metrics. Among these, IoU and $F1$ serve as the primary evaluation criteria, while the remaining metrics serve as supplementary indicators.

C. Implementation Details

Our proposed LGTC model was developed using the PyTorch framework and trained on an NVIDIA RTX A6000 GPU, equipped with 48GB of RAM. During training, the stochastic gradient descent (SGD) optimizer [46] was employed with an initial learning rate of 0.01, weight decay of $1e^{-4}$, and momentum of 0.9. For the experimental setup, the batch size for both labeled and unlabeled training sets was set to 8, and the training process was terminated after 80 epochs. To accelerate model convergence and enhance generalization capability under semi-supervised conditions, we incorporated the pretrained weights of ResNet50 [47] into the LRCBlock of our CD network. In the TCL formulation presented in (2), we set the weighting coefficients to 0.1 for the region-level loss and 0.001 for the image-level loss. In (3), the unsupervised loss coefficient was configured as 1. Additionally, the threshold for generating the confidence mask in (13) was established at 0.95. Notably, the effects of these parameters will be discussed in Section III-F. To enhance the model robustness, we implemented a dual data augmentation strategy during training, comprising both weak and SP schemes. The WP incorporates random flipping, random rescaling within a range of 0.5–2.0, and random cropping. For SP, we adopted nine augmentation techniques from RandAugment [48], including identity, contrast, autocontrast, brightness, color, equalize, sharpness, posterize, and solarize.

D. Performance Comparison

1) Experiments on the LEVIR-CD Dataset: Table I presents the quantitative results of all compared methods on the LEVIR-CD dataset under 5%, 10%, 20%, and 40% labeled ratios. The best values for all performance metrics are highlighted in bold. It can be observed that the proposed LGTC consistently achieves the optimal performance in the three key metrics of $F1$, IoU, and OA across all labeling ratios, demonstrating its strong capability to effectively exploit the latent semantic information in unlabeled data. At the 20% and 40% labeling ratios, LGTC achieves IoU scores exceeding 80%, significantly outperforming a range of comparative methods. Additionally, we conducted control experiments using only different proportions of labeled data for training, denoted as Sup-only. As the labeled ratio decreases, the performance improvement of LGTC over Sup-only becomes more pronounced. For example, at 5%, 10%, and 20% labeling ratios, LGTC improves IoU of 7.62%, 4.75%, and 3.52% than Sup-only, respectively. Moreover, at the 40% labeling ratio, the Sup-only setting already achieves IoU performance comparable to ECPS and surpasses several mainstream semi-supervised methods such as RCL, FPA, and UniMatch. This further

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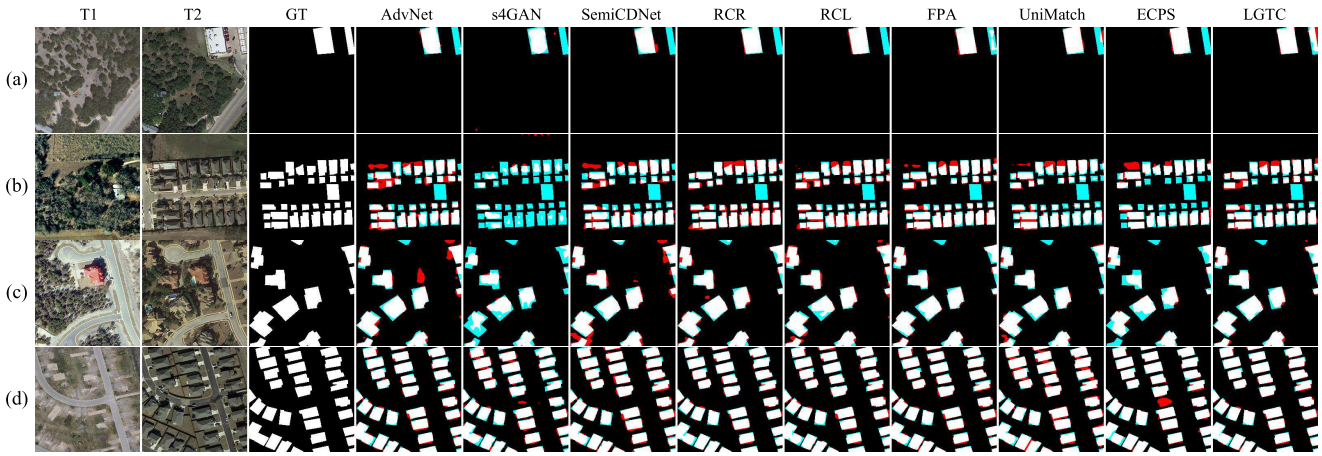


Fig. 4. Qualitative comparison of different methods on the LEVIR-CD dataset. Examples of CD results at the labeled ratio of (a) 5%, (b) 10%, (c) 20%, and (d) 40%.

TABLE II
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE WHU-CD DATASET WITH DIFFERENT LABELED RATIOS

Methods	5%					10%					20%					40%				
	Pre	Rec	IoU	F1	OA	Pre	Rec	IoU	F1	OA	Pre	Rec	IoU	F1	OA	Pre	Rec	IoU	F1	OA
AdvNet	88.47	76.86	69.86	82.26	98.68	84.83	83.82	72.89	84.32	98.76	91.61	82.86	77.02	87.02	99.02	90.67	89.55	82.00	90.11	99.22
s4GAN	88.80	61.45	57.03	72.63	98.16	84.74	75.57	66.52	79.89	98.49	87.98	77.65	70.20	82.49	98.69	90.63	80.63	74.43	85.34	98.90
SemiCDNet	86.32	77.58	69.08	81.72	98.62	88.76	81.76	74.09	85.12	98.87	88.39	84.52	76.08	86.41	98.95	90.52	90.25	82.45	90.38	99.24
RCR	89.98	80.48	73.86	84.97	98.87	92.41	85.16	79.60	88.64	99.13	89.67	86.55	78.70	88.08	99.07	90.90	**91.78**	84.05	91.33	99.31
RCL	**92.38**	77.07	72.46	84.03	98.84	90.20	84.17	77.12	87.08	99.01	92.23	85.13	79.44	88.54	99.13	91.17	91.02	83.65	91.10	99.29
FPA	89.36	83.31	75.79	86.23	98.94	93.97	84.17	79.86	88.80	99.16	93.34	85.62	80.70	89.32	99.19	93.84	90.02	85.00	91.89	99.37
UniMatch	92.32	84.49	78.95	88.24	99.11	**94.10**	83.55	79.39	88.51	99.14	**94.25**	85.72	81.46	89.78	99.23	**93.89**	86.58	81.97	90.09	99.24
ECPS	51.17	81.91	45.97	62.99	96.18	47.48	85.44	43.92	61.04	95.67	56.61	84.96	51.45	67.95	96.82	51.47	91.38	49.08	65.85	96.24
Sup-only	88.30	79.28	71.74	83.55	98.76	87.08	83.34	74.17	85.17	98.85	89.63	85.84	78.09	87.69	99.04	91.32	90.55	83.37	90.93	99.28
LGTC	90.85	**86.55**	**79.61**	**88.65**	**99.12**	92.47	**87.04**	**81.27**	**89.67**	**99.20**	92.74	**88.21**	**82.52**	**90.42**	**99.26**	92.37	91.70	**85.25**	**92.04**	**99.37**

verifies the advantages of our proposed LGTC in change modeling.

Fig. 4 presents the visualized CD results of all methods on the LEVIR-CD dataset with different labeled ratios. To intuitively compare model performance, we denote true positive (TP), true negative (TN), false positive (FP), and false negative (FN) in white, black, red, and blue, respectively. The LEVIR-CD dataset primarily focuses on building additions and removals, where the change regions typically exhibit strong structural characteristics and spatial regularity. In low-label scenarios, models often struggle to accurately identify change regions, resulting in a large number of false positives and FNs. For instance, in Fig. 4(a), most methods fail to detect the rectangular change region in the upper right corner of the image. In Fig. 4(b), the AdvNet, SemiCDNet, and ECPS incorrectly identify the unchanged region in the upper left corner as changed (red areas), indicating that these methods exhibit significant limitations in capturing structural feature representations. Meanwhile, in the absence of sufficient ground-truth supervision, the model tends to exhibit blurred predictions around change boundaries. For example, in Fig. 4(d), due to the dense distribution of buildings, the s4GAN, SemiCDNet, and RCR are unable to clearly delineate the contours of changed objects, and UniMatch suffers from severe region merging within the change areas. In contrast, our method accurately captures spatial continuity and the actual change regions by uniting the multilevel consistency constraints.

2) Experiments on the Whu-CD Dataset: Table II demonstrates the quantitative comparison results of all methods on the WHU-CD dataset under 5%, 10%, 20%, and 40% labeled ratios. At the 10% labeled ratio, the proposed LGTC achieves an IoU score of 81.27%, outperforming the second-ranked FPA by 1.41%. Furthermore, LGTC achieves state-of-the-art performance in both $F1$ and IoU metrics across all four labeling ratios. Specifically, it attains IoU/$F1$-scores of 82.52%/90.42% and 85.25%/92.04% at the 20% and 40% labeling ratios, respectively. These results fully demonstrate LGTC's robustness and superior generalization capability under limited supervision. With the increase in label proportion, the CD performance of the models improves progressively. When the labeling ratio rises from 5% to 10%, UniMatch shows only marginal improvements of 0.44% in IoU and 0.27% in $F1$. In contrast, our Sup-only method gains 2.43%/1.62% in IoU/$F1$, while LGTC improves by 1.66%/1.02%, demonstrating more stable and sustained improvement.

Fig. 5 presents the visual comparison of CD results across all methods under different labeling ratios on the WHU-CD dataset. Similar to LEVIR-CD, WHU-CD also focuses on building change detection. However, it features higher spatial resolution, resulting in larger building scales and more distinct structural contours. In Fig. 5(a), the AdvNet, s4GAN, RCL, and ECPS fail to completely detect the change targets near image edges, whereas our method achieves precise coverage of this region, demonstrating superior sensitivity to boundary details. As shown in

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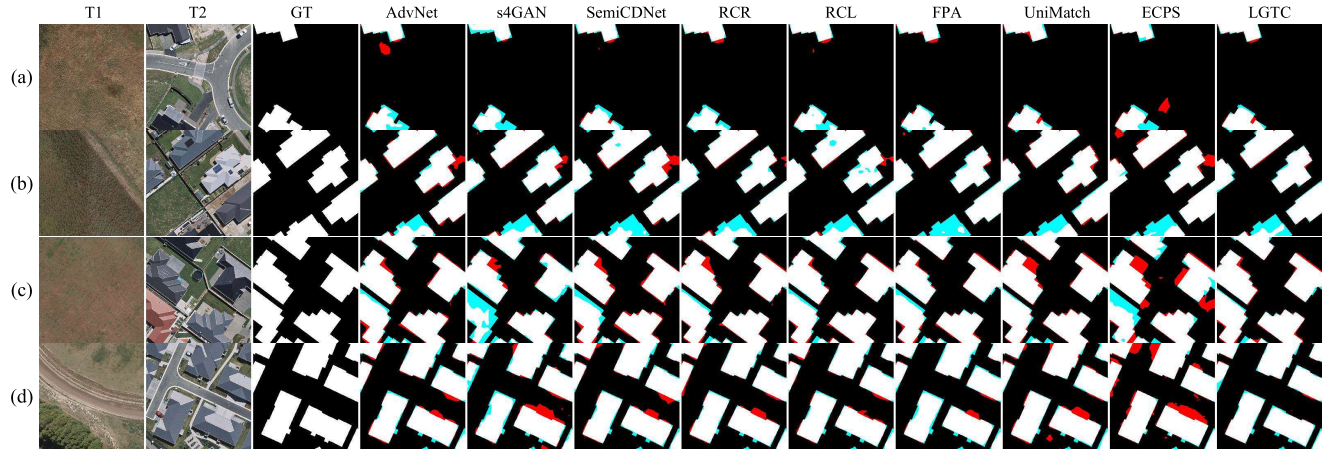


Fig. 5. Qualitative comparison of different methods on the WHU-CD dataset. Examples of CD results at the labeled ratio of (a) 5%, (b) 10%, (c) 20%, and (d) 40%.

TABLE III
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE CDD-CD DATASET WITH DIFFERENT LABELED RATIOS

Methods	5%					10%					20%					40%				
	Pre	Rec	IoU	F1	OA	Pre	Rec	IoU	F1	OA	Pre	Rec	IoU	F1	OA	Pre	Rec	IoU	F1	OA
AdvNet	85.17	73.42	65.10	78.86	94.91	91.90	77.48	72.53	84.08	96.21	91.11	86.33	79.62	88.66	97.14	93.18	88.15	82.81	90.59	97.63
s4GAN	85.11	57.88	52.56	68.90	93.84	91.06	62.41	58.80	74.06	94.84	91.56	73.66	68.97	81.64	96.09	91.32	77.86	72.49	84.05	96.51
SemiCDNet	87.78	69.59	63.44	77.63	94.82	89.93	79.00	72.58	84.11	96.14	92.84	85.22	79.97	88.87	97.24	93.11	88.19	82.79	90.58	97.63
RCR	85.07	70.06	62.39	76.84	94.54	88.87	76.99	70.22	82.51	95.78	90.57	83.15	76.52	86.70	96.70	93.11	87.78	82.43	90.37	97.58
RCL	85.35	**79.90**	70.26	82.53	95.63	88.76	85.58	77.21	87.14	96.74	90.54	**88.84**	81.29	89.68	97.36	91.17	**91.17**	83.78	91.17	97.72
FPA	89.30	76.23	69.85	82.25	95.75	92.66	81.52	76.57	86.73	96.78	92.40	86.10	80.41	89.14	97.29	92.55	86.37	80.75	89.35	97.34
UniMatch	90.53	75.74	69.66	82.49	95.42	92.12	80.95	75.96	85.61	96.27	**93.61**	85.78	79.51	88.60	97.08	93.29	88.39	82.53	90.78	97.34
ECPS	**92.18**	71.87	67.74	80.77	95.75	**93.94**	80.77	76.77	86.86	96.86	92.92	86.44	81.75	89.56	97.50	93.67	87.64	83.43	90.56	96.97
Sup-only	86.33	70.80	63.66	77.80	94.78	89.66	76.90	70.63	82.79	95.87	92.12	82.74	77.27	87.18	96.85	**93.84**	87.42	82.68	90.52	97.63
LGTC	90.93	76.76	**71.30**	**83.25**	**96.01**	93.13	**82.43**	**77.71**	**87.46**	**96.94**	93.41	86.94	**81.92**	**90.06**	**97.52**	93.63	89.08	**83.99**	**91.30**	**97.80**

Fig. 5(b) and (d), the competing methods consistently exhibit erroneous red regions along prediction boundaries, while LGTC maintains more accurate alignment with the ground-truth contours of the changed objects. In Fig. 5(c), the newly constructed building in $T2$ shares a similar surface color with its surroundings, leading to false change detection in RCR, UniMatch, and ECPS. In contrast, LGTC effectively suppresses background interference and accurately localizes the target region. This superior performance benefits from the synergistic design of local–global modeling in the LGIBlock and multilevel constraints in TCL, enabling more accurate discrimination of change regions.

3) Experiments on the CDD-CD Dataset: Table III shows the quantitative experimental results of all methods on the CDD-CD dataset across four labeled ratios. Compared to the previous two datasets, CDD-CD contains more complex change objects with greater morphological diversity, which significantly increases the difficulty of modeling effective change patterns under low-labeling scenarios and leads to performance decline. For instance, at the 5% labeled ratio, only RCL achieves an IoU of 70.26%, and all other compared methods fail to surpass the 70% IoU threshold. In contrast, our method achieves an IoU of 71.30% as the state-of-the-art performance. This demonstrates our approach's superior adaptability to diverse change types. As the labeling ratio increases, all methods exhibit significant performance improvements. However, LGTC consistently achieves optimal results in $F1$, IoU, and OA metrics across all labeling ratios, further validating the superiority of the proposed framework. Notably, at the 40% labeling ratio, our Sup-only approach achieves competitive performance of 82.68% IoU and 90.52% $F1$, surpassing or matching semi-supervised counterparts such as s4GAN, RCR, and FPA. This remarkable performance demonstrates the powerful capabilities of our designed feature extraction backbone in feature modeling.

Fig. 6 presents the CD visualization results of all methods on the CDD-CD dataset under different labeling ratios. As shown in Fig. 6(a), all methods except our proposed LGTC exhibit significant false positives or FNs when identifying irregular change regions. Additionally, the substantial color discrepancy between $T1$ and $T2$ further increases the difficulty of modeling semantic consistency, leading to extensive misdetections by s4GAN, RCL, and UniMatch. In contrast, LGTC effectively captures global semantic correlations and suppresses the interference of irrelevant noise, achieving accurate change identification. As illustrated in Fig. 6(b) and (c), most comparative methods exhibit incomplete detection (blue areas) due to natural variations. In Fig. 6(d), object occlusion significantly degrades boundary recognition capabilities across most methods. However, LGTC achieves more accurate detection results for complex structural changes by leveraging the powerful feature extraction module and effective multilevel consistency learning mechanism to enhance the modeling of spatial contextual relationships.

E. Ablation Study

1) Effectiveness of the LGIBlock: To validate the effectiveness of the proposed LGIBlock and its components, we

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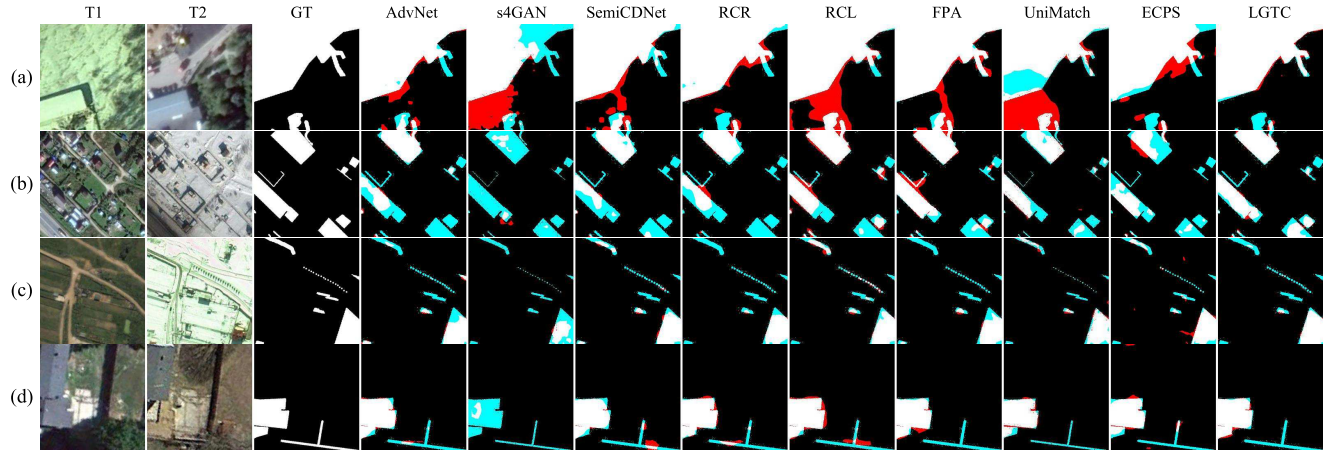


Fig. 6. Qualitative comparison of different methods on the CDD-CD dataset. Examples of CD results at the labeled ratio of (a) 5%, (b) 10%, (c) 20%, and (d) 40%.

TABLE IV

ABLATION STUDY OF THE LGIBLOCK ON THE WHU-CD DATASET WITH A 5% LABELED RATIO

Mode	Encoder	WHU-CD 5%				
		Pre	Rec	IoU	F1	OA
Semi	ResNet18	85.92	75.85	67.47	80.57	98.55
	ResNet34	**91.99**	79.92	74.72	85.53	98.93
	ResNet50	90.01	83.58	76.49	86.68	98.98
	GFTBlock	90.92	72.41	67.53	80.62	98.62
	LGIBlock	90.85	**86.55**	**79.61**	**88.65**	**99.12**
Sup-only	ResNet18	72.88	64.28	51.88	68.31	97.63
	ResNet34	75.82	77.88	62.39	76.84	98.14
	ResNet50	87.05	72.57	65.50	79.15	98.48
	GFTBlock	59.57	72.94	48.79	65.58	96.96
	LGIBlock	**88.30**	**79.28**	**71.74**	**83.55**	**98.76**

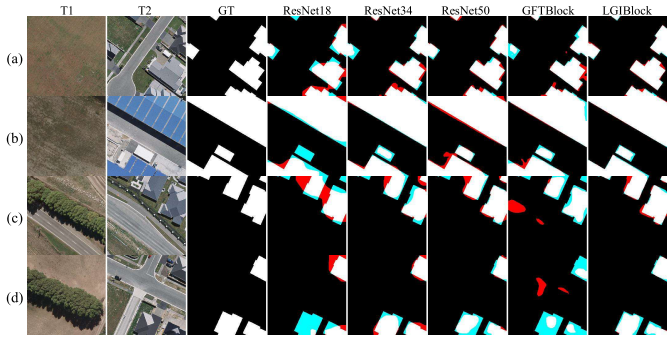


Fig. 7. Ablation study of the LGIBlock on the WHU-CD dataset with 5% labeled ratio under (a) and (b) semi-supervised and (c) and (d) Sup-only settings.

conducted ablation experiments on the WHU-CD dataset using 5% labeled data under both semi-supervised and Sup-only settings. We selected three widely used ResNet variants, ResNet18, ResNet34, and ResNet50, as encoder backbones for comparison, all of which are composed of LRCBlocks. Furthermore, we also constructed an encoder by stacking GFTBlocks for comparative analysis. The experimental results are presented in Table IV. It can be observed that the encoder backbone built with the LGIBlock consistently achieves superior performance under both the semi-supervised mode with TCL and the Sup-only mode. Specifically, in a semi-supervised setting, the LGIBlock achieves IoU improvements of 3.12% compared to ResNet50, while in the Sup-only setting, the gain reaches 6.24%. These significant improvements strongly validate the substantial advantages of LGIBlock's global frequency modeling capability in enhancing change detection accuracy. Compared with the GFTBlock, the LGIBlock achieves an $F1$ gain of 8.03% and 17.97% under semi-supervised and Sup-only conditions, respectively, fully demonstrating that the LGIBlock has a more powerful context modeling capability through the effective local–global feature interaction mechanism.

In addition, we conducted visual analysis on selected samples, as shown in Fig. 7. The comparative visualization reveals that conventional ResNet architectures fail to accurately delineate building contours, exhibiting numerous FP and FN regions at the edges. And the performance in locating changing areas of the model with the GFTBlock deteriorates significantly, resulting in many false detections. In contrast, our LGIBlock demonstrates superior change perception capability through effective fusion of local and global information and suppression of background noise.

2) Effectiveness of TCL: To comprehensively evaluate the effectiveness of each component in TCL, we conducted ablation experiments on the WHU-CD dataset using two encoder backbones, the LGIBlock and ResNet18. The labeled ratio was set to 5%, and the experimental results are presented in Table V. First, Sup-only denotes the supervised baseline model trained exclusively on labeled data. Subsequently, we further introduce pixel-level, region-level, and image-level consistency losses for SSL with unlabeled data, denoted as LPC, LRC, and LIC, respectively. It can be observed that all three constraints contribute to performance improvements. For instance, when using the LGIBlock as the encoder, LPC, LRC, and LIC gain IoU of 6.41%, 0.49%, and 1.56%, respectively, compared to Sup-only. The improvements are even more pronounced with the ResNet18 encoder, reaching 8.84%, 4.56%, and 3.3%, respectively. These results indicate that each

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TABLE V
ABLATION STUDY OF TCL ON THE WHU-CD DATASET WITH A 5% LABELED RATIO

Model	L_l	L_{pc}	L_{rc}	L_{ic}	LGIBlock					ResNet18				
					Pre	Rec	IoU	F1	OA	Pre	Rec	IoU	F1	OA
Sup-only	✓				88.30	79.28	71.74	83.55	98.76	72.88	64.28	51.88	68.31	97.63
LPC	✓	✓			**91.60**	84.19	78.15	87.74	99.07	75.37	75.75	60.72	75.56	98.06
LRC	✓		✓		88.31	79.86	72.23	83.88	98.78	76.14	68.56	56.44	72.15	97.90
LIC	✓			✓	88.87	80.71	73.30	84.59	98.83	77.54	65.68	55.18	71.12	97.88
LPRC	✓	✓	✓		93.28	83.04	78.36	87.86	99.09	83.21	72.15	62.98	77.29	98.32
LPIC	✓	✓		✓	91.45	85.62	79.27	88.44	99.11	84.37	73.01	64.31	78.28	98.39
LRIC	✓		✓	✓	86.21	79.54	70.56	82.74	98.68	76.89	63.97	53.65	69.84	97.81
TCL	✓	✓	✓	✓	90.85	**86.55**	**79.61**	**88.65**	**99.12**	**85.92**	**75.85**	**67.47**	**80.57**	**98.55**

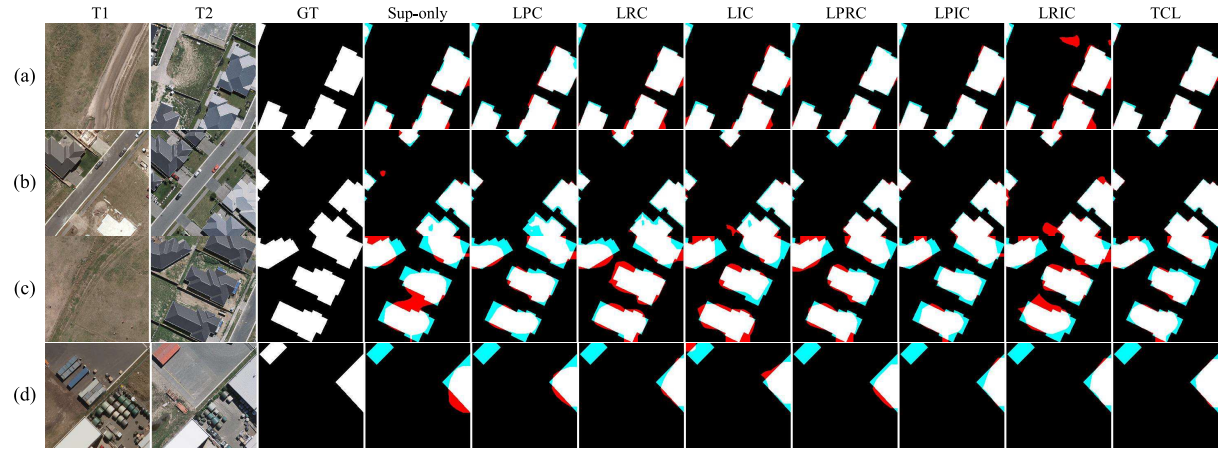


Fig. 8. Ablation study of TCL on the WHU-CD dataset with a 5% labeled ratio using (a) and (b) LGIBlock and (c) and (d) ResNet18 as the encoder.

level of consistency learning effectively exploits latent change information from unlabeled data.

Next, based on the Sup-only baseline, we investigated pairwise combinations of consistency constraints to analyze their synergistic effects. The results demonstrate that combining L_{pc} with either L_{rc} or L_{ic} (LPRC or LPIC) consistently outperforms individual losses. For example, with the LGIBlock encoder, LPRC outperforms LRC by 6.13% in IoU, and LPIC surpasses LIC by 5.97%. However, when L_{rc} and L_{ic} are combined (LRIC), performance drops significantly, suggesting potential feature interference or coupling difficulty between the two. In contrast, our proposed TCL, which successfully integrates all three consistency supervision, achieves the best performance across both encoder architectures. This confirms the effective synergy between multilevel consistency strategies and the architectural superiority of the unified design.

Additionally, Fig. 8 presents the visual comparison of ablation results. It demonstrates that the removal of any individual consistency loss in TCL leads to measurable performance degradation, manifested by an increase in red and blue areas. This performance drop confirms that each constraint level contributes uniquely to the overall effect. In contrast, the complete TCL framework maintains optimal performance, which can effectively and synergistically guide the model to focus on genuine change regions by establishing multilevel consistency constraints.

F. Parameters Analysis

In our model, several key hyperparameters require manual configuration, including the region-level loss weight α, image-level loss weight β, unsupervised loss weight λ, and confidence mask threshold τ. To investigate the impact of different parameter settings on CD performance, we conducted experiments on the WHU-CD dataset using 10% labeled data, as shown in Table VI. For each parameter, we established two comparative configurations by increasing and decreasing the value based on our recommended setting to verify the rationality of the selected configuration. It is worth noting that the parameter settings in this article are determined through experimental evaluation and have demonstrated good performance in our experiments.

As illustrated in Table VI, compared to the setting of 0.1 adopted in this article, CD performance deteriorates when the

TABLE VI
PARAMETERS ANALYSIS ON THE WHU-CD DATASET WITH 10% LABELED RATIO

Parameter	Value	WHU-CD 10%				
		Pre	Rec	IoU	F1	OA
α	0.01	90.53	**88.05**	80.62	89.27	99.16
	0.1	92.47	87.04	**81.27**	**89.67**	**99.20**
	0.5	**92.48**	86.84	81.11	89.57	**99.20**
β	0.0001	**92.57**	86.79	81.14	89.59	**99.20**
	0.001	92.47	**87.04**	**81.27**	**89.67**	**99.20**
	0.05	91.65	86.95	80.57	89.24	99.17
λ	0.3	93.04	86.22	81.00	89.50	99.20
	1	92.47	**87.04**	**81.27**	**89.67**	99.20
	1.7	**93.33**	86.17	81.17	89.61	**99.21**
τ	0.8	90.14	**87.43**	79.80	88.76	99.12
	0.95	**92.47**	87.04	**81.27**	**89.67**	**99.20**
	0.99	91.47	86.11	79.71	88.71	99.13

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region-level loss weight α is set to 0.01 or 0.5, with IoU decreasing by 0.65% and 0.16%, respectively. This indicates that a lower weight may fail to effectively leverage regional contextual information to guide feature distribution learning, while an excessively high weight may cause excessive reliance on pseudo-labels to amplify error propagation from noisy labels, both of which are detrimental to performance improvement. For the image-level loss weight β, compared to our baseline setting of 0.001, when reduced to 0.0001, the insufficient weighting causes degradation in the reconstruction branch. This weakens the model's ability to compensate for the limitations of pseudo-label supervision, leading to performance declines of 0.13% in IoU and 0.08% in $F1$. Conversely, increasing the weight to 0.05 overemphasizes the reconstruction task, causing deviation from the primary optimization objective and leading to a reduction of 0.7% in IoU and 0.43% in $F1$. It is important to note that these two loss weights are defined relative to the pixel-level loss scale.

The unsupervised loss weight plays a pivotal role in guiding the semi-supervised model to extract meaningful information from unlabeled data. Excessive weighting may induce overfitting to noisy pseudo-supervision signals, thereby amplifying label errors and degrading detection accuracy. Conversely, insufficient weighting critically undermines the model's capacity to leverage the abundant unlabeled data. Experimental results show that setting the weight λ to 0.3 and 1.7 leads to IoU drops of 0.27% and 0.1%, respectively, compared to the value of 1. Accordingly, we set the weight to 1 in this article to achieve a desirable balance between stability and performance, while preserving the scale consistency between the unsupervised and supervised loss terms.

Experimental results demonstrate that a confidence mask threshold of 0.95 yields optimal performance. Reducing the threshold to 0.8 results in 1.47% IoU and 0.91% $F1$ decrease, primarily attributed to the inclusion of excessive unreliable pixels in consistency training, which accumulates noise and reinforces erroneous predictions. On the other hand, increasing the threshold to 0.99 causes 1.56% IoU and 0.96% $F1$ reduction, as excessively stringent sample selection critically limits the learning capacity of the semi-supervised framework. Therefore, a threshold of 0.95 is adopted, allowing only high-confidence pixels to contribute to the unsupervised loss computation, which helps maintain training stability.

IV. Conclusion

This article proposes a novel SSCD framework, LGTC, to fully exploit the potential of unlabeled data and enhance the ability to capture the structural information of change regions and noise robustness. LGTC introduces the TCL strategy, which implements a hierarchical supervision mechanism through pixel-, region-, and image-level consistency constraints, effectively improving SSL performance. Furthermore, the LGIBlock is incorporated into the feature extraction network to effectively capture both fine-grained details and global change patterns of change regions, while reducing the impact of noise and significantly enhancing multiscale change perception capabilities. Extensive experiments on three public datasets demonstrate that LGTC achieves superior performance, validating its effectiveness. Although semi-supervised CD methods have made substantial progress under label-scarce conditions, they remain fundamentally dependent on a certain amount of high-quality annotations. In future work, we plan to explore fully unsupervised CD models to eliminate annotation dependency entirely, thereby improving the scalability of change detection in real-world applications.

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